



Agenda

- ① Introduce Map Reduce Framework
 - ② Different phases in Mapper and Reducer
 - ③ How is a program executed in Map Reduce
 - ④ Explaining code of Map Reduce
-

Data Modelling - Problem Statement

- ① OLTP vs OLAP in usecase
- ② Introduction to Data Warehousing
- ③ Benefits of Data Warehousing
- ④ Designing Data Warehouse

Map - Reduce Framework

Programming model for processing large distributed data sets

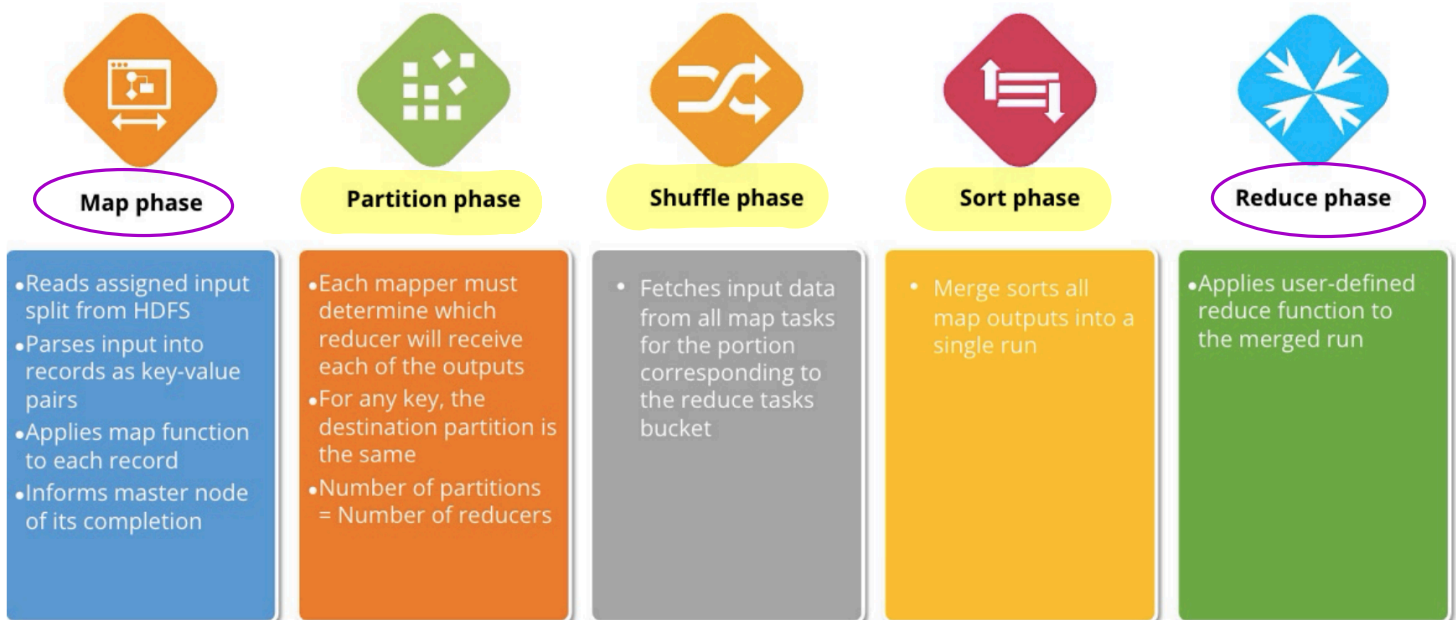
Users specify map-function

takes input from distributed machine and generates an intermediate $\langle \text{key}, \text{value} \rangle$ pair

and reduce function

that merges all values associated with the same intermediate key

Different Phases in Mapper and Reducer



Mapper
Logic

Reducer
Logic

Responsibility of developer

- ① Map function
- ② Reduce

Provided by Map Red Framework

- ① partition
- ② shuffle
- ③ sort

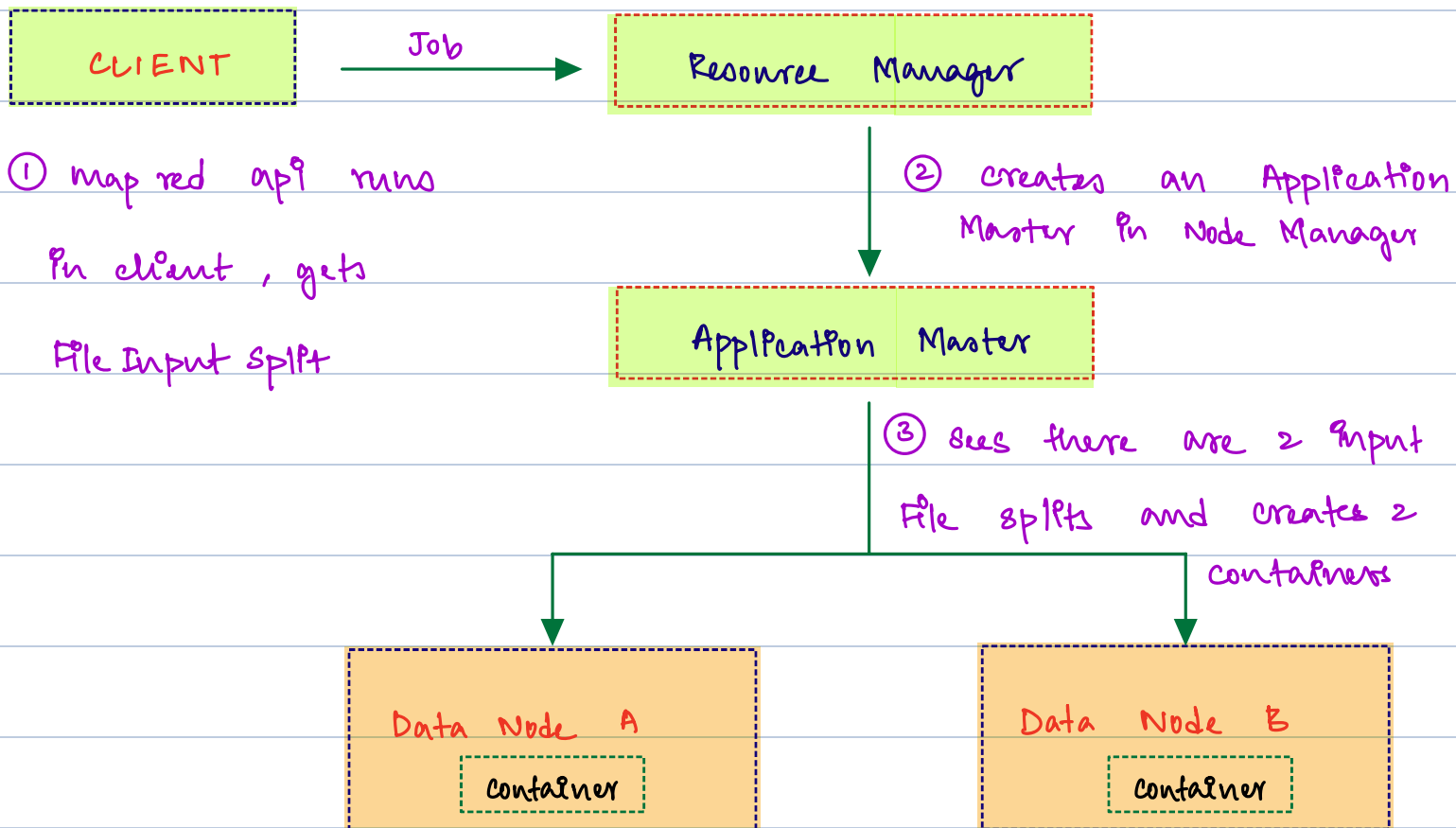
Mapper Logic

Input File Size = 256 MB

No. of reducers configured = 1

Constraints

256 MB → 2 File Blocks, present in Data Node A and B



Map Phase

④ Map tasks starts (In each of the Data Node)

① Reads and executes `mapper.py`

② Runs partitions (inbuilt logic)

Creates 1 partition for each mapper as

no. of reducers configured = 1

Intermediate output

⑤ Notify Application manager Informing Mapper as completed

Mapper Code

```
#!/usr/bin/env python

import sys
import re

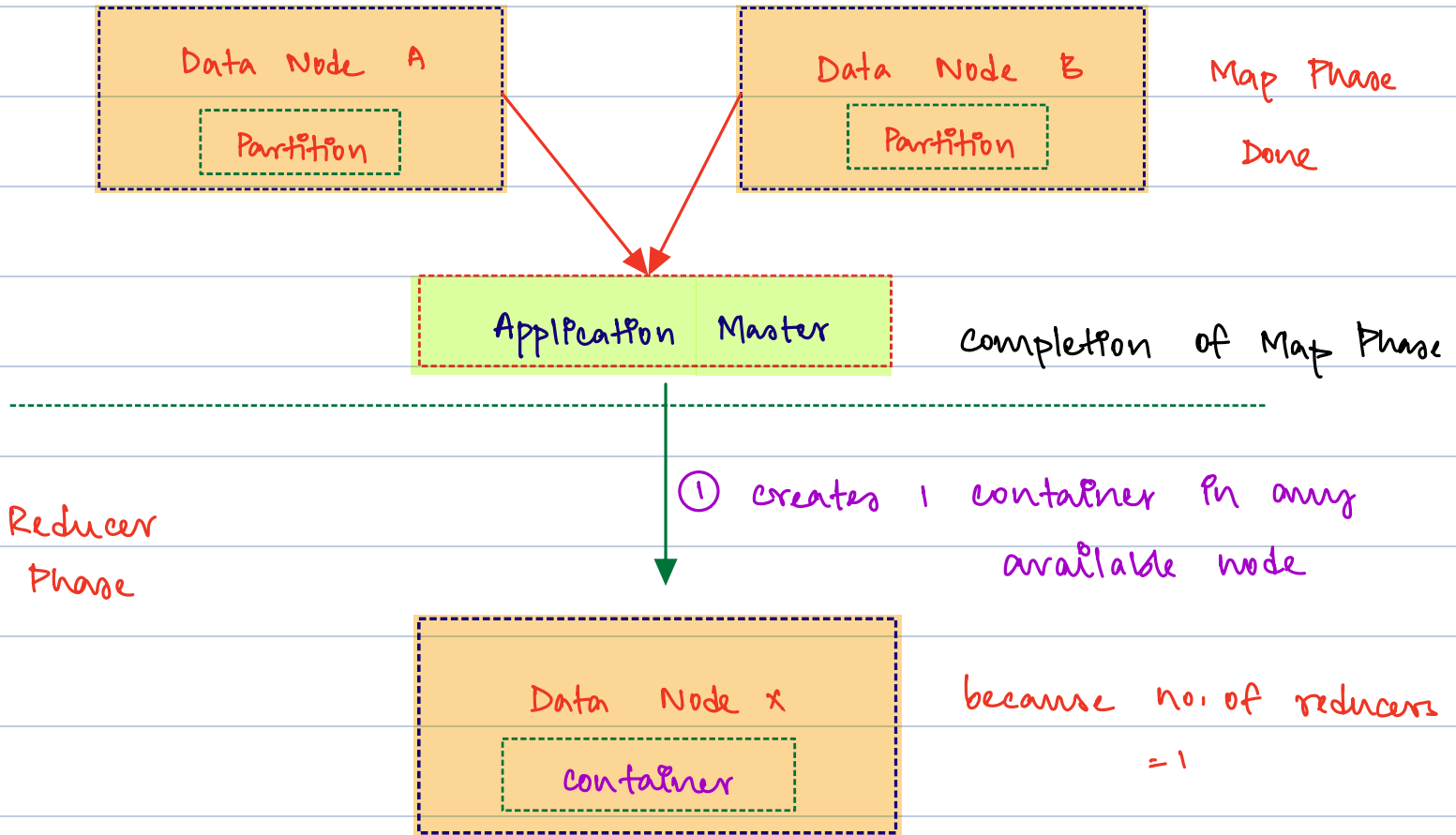
log_pattern = re.compile(r'[\d{2}/\w{3}/\d{4}:(\d{2}):(\d{2}):(\d{2}).*?\s]')

for line in sys.stdin:
    line = line.strip()
    match = log_pattern.search(line)
    if match:
        extracted_hour = match.group(1)
        print('{}\t{}'.format(extracted_hour, 1))
```

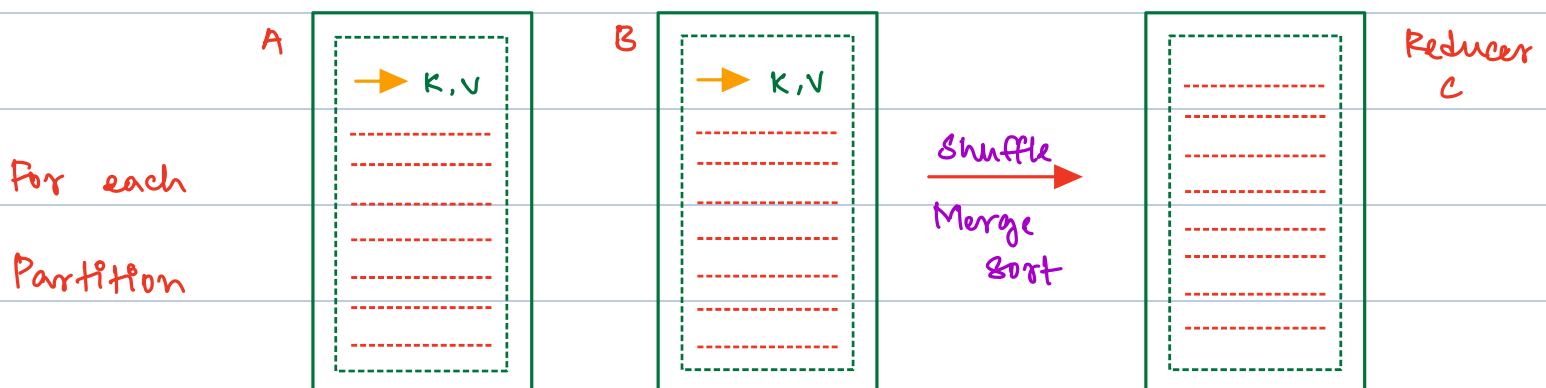
Reducer logic

No. of reducers configured = 1

Constraints



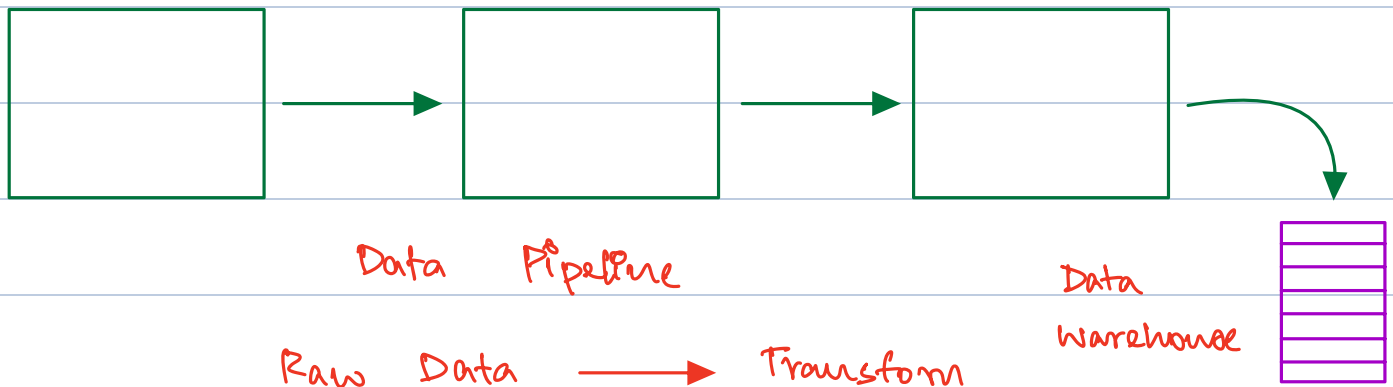
② Reducer container reads partition 0 from Data Node A and Data Node B



- ③ For each Key, group the values
- ④ Execute the reducer logic on the Key
- ⑤ Write the final result to HDFS output path
- ⑥ Inform Application Master about completion
- ⑦ Inform Resource Manager and release the containers

Reducer Code

Data Engineer → write Pipelines



Question

- ① What are the top 10 books sold weekly? For last 10 Years
- ② What are the other products frequently bought with books?
- ③ Who are the top authors based on book sales?

OLTP

online transaction processing
For running small concurrent transactions

→ add new Books,
customer, purchase

OLAP

online analytical processing
For running analytic query

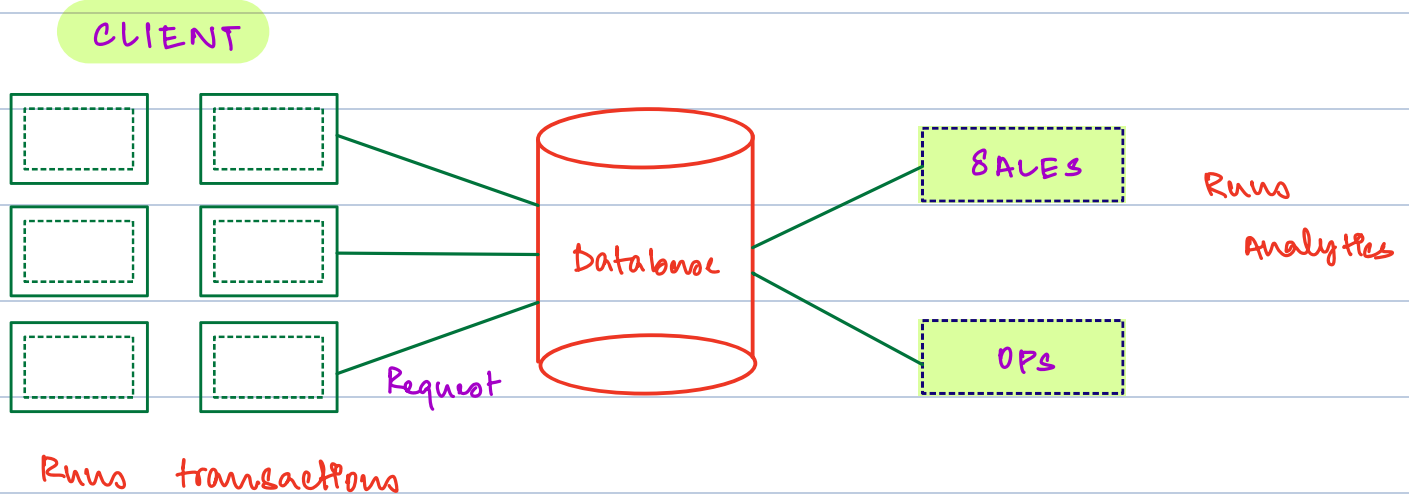
→ Revenue from
The Intelligent Investor
over last 20 years per
region

OLTP

RDBMS System → MySQL, MSSQL, SQLite, Oracle, PostgreSQL

Queries maintains ACID properties

CRUD operations → Insert, retrieve, update and delete



Resource Crunch

analytical queries takes more resources to execute

Solution

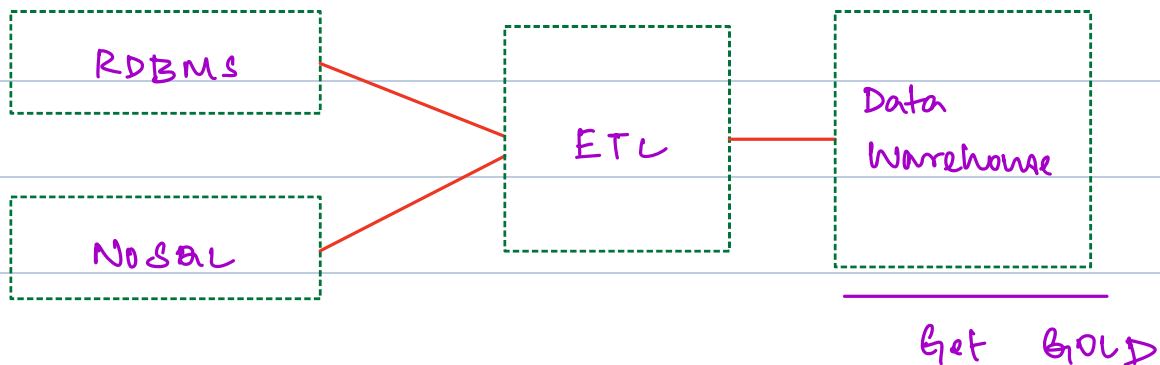
Move to a dedicated system to handle analytical query.

Introduction to data warehousing

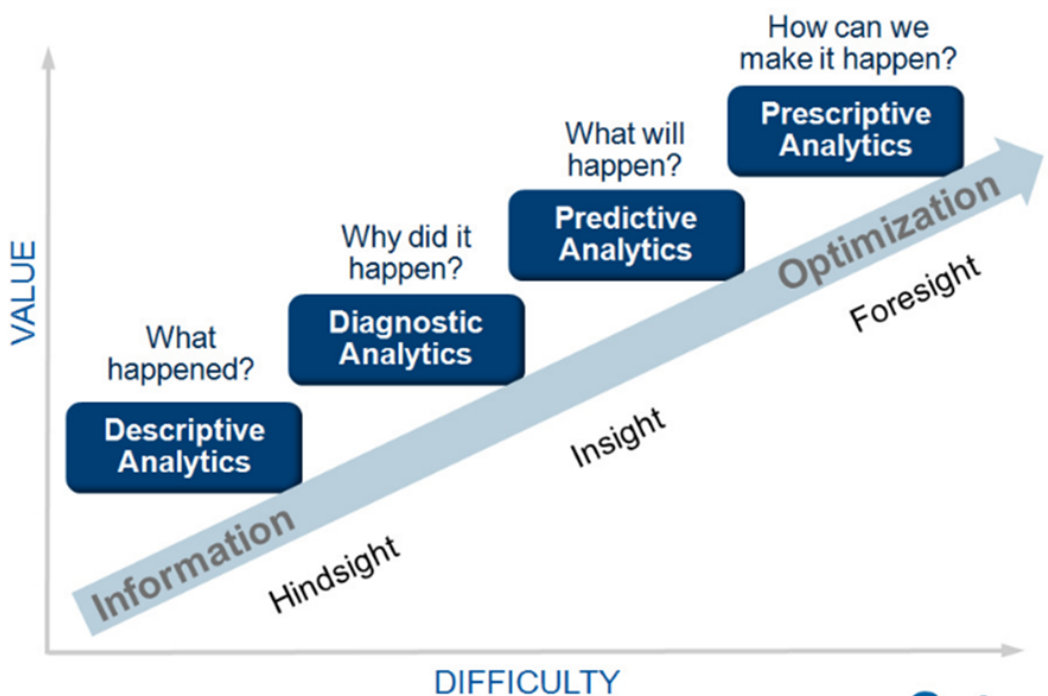
Storage unit to support business analytics and reporting.

Typically a data warehouse contains a large volume of current and historical data.

How is data moved to Data warehousing?



Analytic Value Escalator



Analytics
on top of
Data Warehouses

Designing Data Warehouse

The design of warehouse completely depends on the user requirement.

Dimension Modelling

- ① What are the top 10 books sold weekly? For last 10 Years
- ② What are the other products frequently bought with books?
- ③ Who are the top authors based on book sales?

Facts (Measurements (Metric))

- ① Quantity of Books sold
- ② Price of Individual Books

Dimensions (Who, When, What, Where)

- ① Author
- ② Name of Book
- ③ Category
- ④ Country / City
- ⑤ Date, time, year, week
- ⑥ Other products

Stop